OR

2) Write a program to read the contents of "abc.txt" file, Display the contents of file in uppercase as output.

Step 1: vim abc.txt

Add text data in this txt file

Step 2: vim FileUpper.java

```
import java.io.*;
public class FileUpper{
public static void main(String[] args){
try{
FileReader fr=new FileReader("abc.txt");
int ch;
while((ch=fr.read())!=-1){
System.out.print(Character.toUpperCase((char)ch));
}
fr.close();
}
catch(IOException e){
System.out.println(e);
}
}
}
```

Slip 15

1) Write a Java program to reverse a number. Accept the number using command line arguments.

```
public class ReverseNumber{
public static void main(String[] args){
int number;
```

```
int reverse=0;
int remainder;
if(args.length==0){
System.out.println("Please enter a number using command line argument");
return;
}
number=Integer.parseInt(args[0]);
System.out.println("Original Number = "+number);
while(number>0){
remainder=number%10;
reverse=reverse*10+remainder;
number=number/10;
}
System.out.println("Reversed Number = "+reverse);
}
}
```

2) Create a class circle (member – radius), cylinder (members – radius, height) which implements this interface. Calculate and display the area and volume.

```
import java.util.Scanner;

interface Shape{
void calculate();
void display();
}

class Circle implements Shape{
double radius;
```

```
double area;

Circle(){
radius=0;
}

Circle(double radius){
this.radius=radius;
}

public void calculate(){
area=3.14*radius*radius;
}

public void display(){
System.out.println("Radius of Circle = "+radius);
System.out.println("Area of Circle = "+area);
}
}

class Cylinder implements Shape{
double radius;
double height;
double area;
double volume;

Cylinder(){
radius=0;
```

```
height=0;
}

Cylinder(double radius,double height){
this.radius=radius;
this.height=height;
}

public void calculate(){
area=2*3.14*radius*(radius+height);
volume=3.14*radius*radius*height;
}

public void display(){
System.out.println("Radius of Cylinder = "+radius);
System.out.println("Height of Cylinder = "+height);
System.out.println("Area of Cylinder = "+area);
System.out.println("Volume of Cylinder = "+volume);
}
}

public class ShapeDemo{
public static void main(String[] args){
Scanner sc=new Scanner(System.in);

System.out.print("Enter radius of circle: ");
double r1=sc.nextDouble();
```

```
Circle c=new Circle(r1);
c.calculate();
c.display();
System.out.print("Enter radius of cylinder: ");
double r2=sc.nextDouble()
System.out.print("Enter height of cylinder: ");
double h=sc.nextDouble();
Cylinder cy=new Cylinder(r2,h);
cy.calculate();
cy.display();
}
}
```

OR

2) Write a java program to create using Runnable interface to display traffic signal.

```
class TrafficSignal implements Runnable{
public void run(){
try{
System.out.println("Traffic Signal");
System.out.println("RED - STOP");
Thread.sleep(2000);
System.out.println("YELLOW - READY");
Thread.sleep(2000);
System.out.println("GREEN - GO");
Thread.sleep(2000);
}
catch(InterruptedException e){
System.out.println(e);
```